

# Optics Letters

## Dual-periodically poled lithium niobate microcavities supporting multiple coupled parametric processes

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**Periodically poled lithium niobate (PPLN) microcavities with additional reciprocal vectors attract attention as a platform for efficient parametric nonlinear optical processes with wavelength and polarization flexibility. Here, we report the simultaneous realization of multiple coupled parametric processes in PPLN microdisk cavities with dual periods as a result of the significantly increased number of reciprocal vectors to fulfill quasi-phase matchings for a series of nonlinear processes. PPLN microdisks with up to  $1.43 \times 10^5$  quality factors and unit domain size of 90 nm in width were fabricated using CMOS compatible microfabrication techniques and electrically poled with the help of piezoresponse force microscopy. The conversion efficiency of second-harmonic signal was measured to be  $51\% \text{ W}^{-1}$ . Our work paves the way towards efficient cascaded parametric effects including third- and fourth-harmonic generations.** © 2020 Optical Society of America

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Whispering gallery mode microcavities with high quality ( $Q$ ) factors have the ability to tightly confine light via total internal reflection [1]. They can dramatically enhance light intensity in the whole material transparent window, providing a platform to realize nonlinear optical effects and their applications under a weak pump. Lithium niobate (LN) microcavities, combining the excellent optical properties of LN crystal and the features of microcavities, are very suitable for nonlinear optical investigations. Both second-order and third-order nonlinear optical effects, such as second-harmonic generation (SHG) [2–16], sum-frequency generation [14,17–19], parametric down conversion [15,20–22], Raman effect [16,23,24], and four-wave mixing (FWM) effect [16], were demonstrated in LN microdisks and microrings.

It is known that to obtain continuous growth of a nonlinear optical signal, the phase matching condition should be satisfied

in parametric nonlinear optical processes. In LN microcavities, phase matching was initially realized by manipulating the cavity mode dispersion while considering the birefringent properties of LN crystal [5,6]. Recently, quasi-phase matching was also demonstrated by introducing domain structures into LN microdisk and microring cavities [2,3,8,9,11–13,17,22]. Periodically poled LN (PPLN) structures afford additional wave vectors to compensate for the original mismatch of wave vectors between the interacting light waves, leading to flexibility in terms of wavelengths and polarizations. Consequently, the largest nonlinear optical coefficient  $d_{33}$  of LN can be utilized, which helps to improve the nonlinear conversion efficiency. SHG efficiency has been raised to  $230,000\% \text{ W}^{-1}$  in a racetrack ring cavity by partially using  $d_{33}$  on its arms [11]. It is worth to mention that  $d_{33}$  was also used in  $x$ -cut monocrystalline LN microcavities, where efficient SHG and third-harmonic generation (THG) with  $9.9\% \text{ mW}^{-1}$  and  $1.05\% \text{ mW}^{-2}$  conversion efficiencies, respectively, were observed [10].

Focusing on the domain structures on on-chip LN cavities, in 2018, straight-strip-shaped domain structures with  $23 \mu\text{m}$  [9] and  $16 \mu\text{m}$  spatial periods [8] were introduced into circular microring and microdisk cavities fabricated from PPLN films of corresponding domain structures. In these cavities, chirped domain structures with reciprocal vectors, unfortunately lacking freedom of control, were actually experienced by the whispering gallery modes propagating along the cavity circumference. Later, radial domain patterns with a single spatial period were fabricated on a disc LN cavity [13] and a microring cavity exhibiting  $250,000\% \text{ W}^{-1}$  SHG conversion efficiency [12]. In these radial PPLN cavities, only a single reciprocal vector and its high-order reciprocal vectors that are odd times of the fundamental one can be employed to satisfy the quasi-phase matching condition if the PPLN structure is uniform. A single spatial period was also experienced in a racetrack LN cavity with straight-strip-shaped periodically poled structures on its arms [11].

Drawing on the experience in bulk ferroelectric materials, devices with complex periodic polarized sequences, such as dual-period [25], Fibonacci optical superlattice [26], and aperiodic Thue–Morse sequence [27], support designed multiple reciprocal vectors that cannot survive or are not strong in single-period and chirped periodically polarized structures. Cascaded nonlinear optical effects with high conversion efficiency were demonstrated [25,26] by simultaneously fulfilling quasi-phase matching for multiple parametric nonlinear optical processes with one periodically polarized structure supporting multiple relatively strong reciprocal vectors rather than two separated structures. Multiple parametric coupling processes were implemented in dual-periodic, quasi-periodic, and other aperiodic bulk periodically polarized devices. For example, THG with  $0.26\% \text{ W}^{-2}$  conversion efficiency was realized in a dual-periodic periodically polarized lithium tantalate [25]. PPLN microcavities with designed complex domain structures are promising to realize efficient cascaded nonlinear optical effects under a weak pump.

Recently, we developed a technique to fabricate high- $Q$  PPLN microcavities with complex domain patterns of unit size down to  $\sim 100 \text{ nm}$  based on CMOS compatible micro-fabrication methods and a piezoresponse force microscopy (PFM) poling technique [13]. In this Letter, we report the first preparation of dual-PPLN (DPPLN) microcavities, as far as we know, with  $Q$  factors up to  $1.43 \times 10^5$ . Benefiting from the multiple reciprocal vectors provided by such a DPPLN microdisk, multiple parametric nonlinear optical effects, from SHG to fourth-harmonic generation (FHG), were observed under  $1550\text{-nm}$  continuous pump of several mW power.

Quasi-phase matching can be well understood in the wave vector space. Therefore, it is central to analyze the reciprocal vectors that can be provided by a given PPLN pattern by Fourier decomposition. Taking DPPLN as an example, the two-period PPLN structure can be considered as one periodic  $d_{\text{eff}}$  phase sequence superimposed on another one with a different period in space [25]. Assume the DPPLN consists of two single-period PPLNs defined as  $f_1(x)$  and  $f_2(x)$ , where  $x$  is the light propagation direction. In a whispering gallery mode cavity, it is the tangential direction of the cavity circumference. The Fourier series expression of  $f_1(x)$  and  $f_2(x)$  can be written as

$$f_1(x) = \sum_{m=-\infty}^{\infty} g_m e^{-iG_m x}, \quad f_2(x) = \sum_{n=-\infty}^{\infty} g_n e^{-iG_n x}, \quad (1)$$

respectively, where  $m$  and  $n$  are integers indicating Fourier orders. Here,  $G_m = 2m\pi/\Lambda_1$  and  $g_m = 2 \sin(m\pi D_1)/m\pi$  are the reciprocal vectors and their corresponding Fourier expansion coefficient of PPLN of  $f_1(x)$ , respectively. Similarly,  $G_n = 2n\pi/\Lambda_2$  and  $g_n = 2 \sin(n\pi D_2)/n\pi$  are the parameters for PPLN of  $f_2(x)$ . Among them,  $\Lambda_{1,2}$  and  $D_{1,2}$  are the spatial period and duty cycle of  $f_{1,2}(x)$ , respectively. Accordingly, the DPPLN with a sign function written as  $f(x) = f_1(x)f_2(x)$  is of the Fourier series

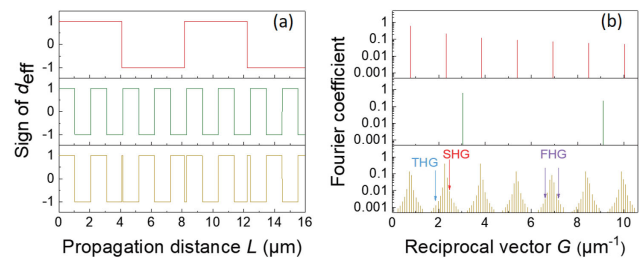
$$f(x) = \sum_{m,n=-\infty}^{\infty} g_m g_n e^{-i(G_m+G_n)x} = \sum_{m,n=-\infty}^{\infty} g_{m,n} e^{-iG_{m,n}x}. \quad (2)$$

The reciprocal vectors of the DPPLN  $G_{m,n} = G_m + G_n$  are the linear superposition of the wave vectors supported by the

single-period PPLN that constructs the DPPLN, while the Fourier expansion coefficient  $g_{m,n} = g_m g_n$  is the product of their Fourier coefficients. Both  $G_{m,n}$  and  $g_{m,n}$  are determined by two separated Fourier orders labeled as  $m$  and  $n$ . By choosing suitable  $m$  and  $n$  and optimizing  $\Lambda_1$  and  $\Lambda_2$  of DPPLN, one is able to obtain multiple reciprocal vectors  $G_{m,n}$  with relatively large Fourier expansion coefficients  $g_{m,n}$  enabling efficient cascaded parametric nonlinear processes by satisfying multiple quasi-phase matching conditions at the same time.

Figure 1(a) represents the schematic of two PPLNs with  $\Lambda_1 = 8.15 \mu\text{m}$  (red line) and  $\Lambda_2 = 2.07 \mu\text{m}$  (green line) and the superposition of them (yellow line) from top to bottom, respectively. The PPLN structures are demonstrated by the change of the sign of effective nonlinear coefficient with the propagation distance in each periodic polarization structure. In the DPPLN structure, the minimum domain unit has a  $69\text{-nm}$  length. This DPPLN was designed to simultaneously achieve SHG of  $1550\text{-nm}$  pump and the sum-frequency generation of the pump and its SHG signal in a microdisk with a  $20\text{-}\mu\text{m}$  radius. To clearly demonstrate the characteristics of the reciprocal vectors of a DPPLN, the Fourier coefficients of the reciprocal vectors provided by the domain structures in Fig. 1(a) were obtained by performing Fourier series expansion, and are correspondingly shown in Fig. 1(b). By comparing the yellow line related to DPPLN with the red and green lines corresponding to PPLN with periods of  $\Lambda_1$  and  $\Lambda_2$ , respectively, we can see that the DPPLN structure supports many more reciprocal vectors. Two wave vectors with Fourier expansion coefficients larger than  $0.4$ , indicating multiple vectors with relatively large Fourier expansion coefficients, are supported. If the center of the domain pattern of the DPPLN deviates from that of the LN microdisk, more reciprocal vectors appear while approximately keeping the envelope of the Fourier spectrum [13].

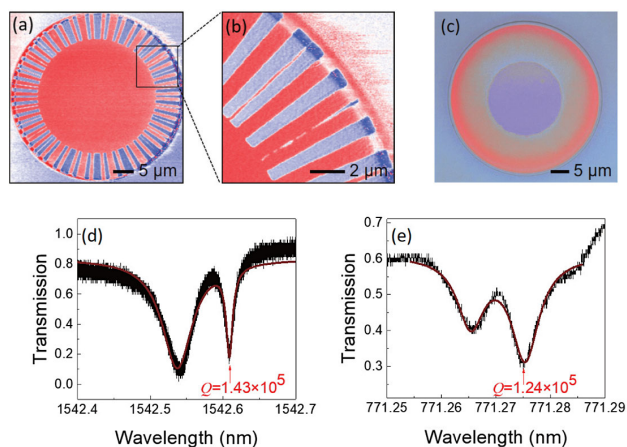
We use a  $z$ -cut LN-on-insulator (LNOI) wafer supplied by NANOLN in Jinan, China, to fabricate DPPLN microcavities. The LNOI wafer consists of a  $700\text{-nm}$  LN film, a  $100\text{-nm}$  chromium (Cr), a  $2\text{-}\mu\text{m}$  silica sacrificial layer, and a  $0.5\text{-mm}$  LN substrate from top to bottom. The preparation steps of the DPPLN microcavities are almost the same as in our previous paper [13]. First, a layer of photoresist was spin-coated on the upper surface of the wafer for ultra-violet (UV) lithography. Then, a photoresist mask of the desired structure for the subsequent argon ion plasma etching was obtained by UV irradiation and following chemical developing. Argon ion plasma etching removed the exposed LN film and thus transferred the photoresist pattern to the LN film. The residual photoresist was cleaned with acetone so that a circular LN pad was finally formed. The next most important step is to periodically polarize



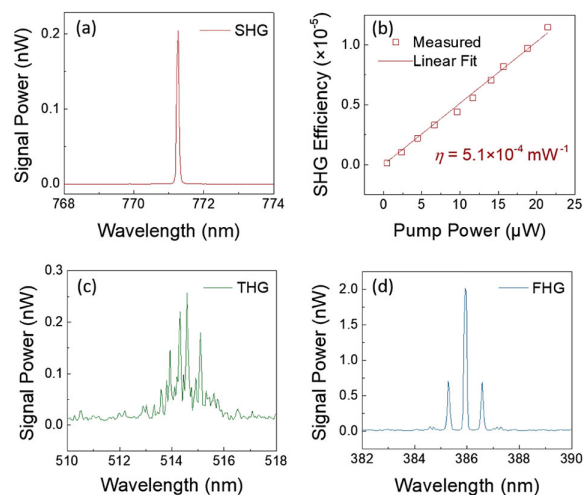
**Fig. 1.** (a) Schematic diagram of the PPLN structure indicated by the sign of  $d_{\text{eff}}$ . The red, green, and yellow lines indicate PPLN with  $\Lambda_1 = 8.15 \mu\text{m}$ , PPLN with  $\Lambda_2 = 2.07 \mu\text{m}$ , and their superposition, respectively. (b) Reciprocal vectors corresponding to (a).

the LN disk by applying an electric field over the LN coercive field of  $\sim 21$  kV/mm with the help of PFM [28]. By scanning the voltage applied PFM tip point by point, a domain pattern with  $512 \times 512$  points can be formed like a drawing in less than 10 min. The size of the PPLN pattern is limited by the moving range of the linear stage to control the position of the PFM tip, which is  $90 \times 90 \mu\text{m}^2$  in our experiments. Finally, we made an edge-suspended DPPLN microdisk by removing the Cr and silica that were originally exposed and covered by the edge of the LN disk.

The domain pattern of the DPPLN disk could be *in situ* characterized just after the LN disk was polarized by using the PFM. Figure 2(a) presents the PFM image of a typical DPPLN microdisk cavity, where the polarization reversed and intact regions are marked by blue and red colors, respectively. Figure 2(b) shows the details of the domain pattern within the black square in Fig. 2(a), which has a small domain unit of 90-nm width. As far as we know, this is the smallest domain unit that has been processed in a periodically polarized microcavity. Additionally, it is worth mentioning that the domain structures fabricated following the aforementioned method were confirmed to have a stability of more than one year, enabling practical applications concerning device stability. Figure 2(c) is an optical micrograph of a final DPPLN microdisk cavity with a silica pillar formed by HF etching. The top surface of the silica pillar is indicated by the inner purple circle, while its side wall is presented by the color graded ring just out of the inner red circle. The gray ring surrounding the outer red circle is the tilt edge of the 20- $\mu\text{m}$ -radius PPLN disk formed during the argon ion etching process. The transmission spectra near 1550 nm and 775 nm for the tapered-fiber-coupled DPPLN disk were, respectively, collected by an optical diode while scanning the wavelengths of the laser sources. By fitting the transmission peaks with the Lorentzian function, we obtained the measured  $Q$  factors of the fabricated microcavities, which were  $1.43 \times 10^5$  at 1542.6 nm



**Fig. 2.** Characterizations of DPPLN microdisks. (a) PFM image of DPPLN microdisk without post-processing; the blue area at the edge of the microdisk is the one that has been polarized. (b) Details of the polarized domain structure marked in black box of (a). The minimum width of the domain unit is measured to be 90 nm. (c) Optical microscope image of 20- $\mu\text{m}$ -radius DPPLN microdisk with its edge suspended in midair; the central purple area represents the silica pillar. (d), (e) Measured  $Q$  factors of the DPPLN microdisk resonator. The loaded  $Q$  factors obtained by Lorentzian fit are  $1.43 \times 10^5$  at 1542.6 nm and  $1.24 \times 10^5$  at 771.3 nm.



**Fig. 3.** Nonlinear signals in dual-periodically poled microdisk resonator. (a) SHG signal with pump power of 0.02 mW. (b) Normalized SHG conversion efficiency, measured to be  $51\% \text{ W}^{-1}$  by linear fitting. (c), (d) Third- and fourth-harmonic signals with pump power of 6.65 mW, respectively.

and  $1.24 \times 10^5$  at 771.3 nm. High  $Q$  factors and small mode volumes of on-chip microcavities help us to get nonlinear optical effects with high efficiency under a weak pump.

Nonlinear optical experiments in DPPLN microdisk cavities were performed using the same experimental setup as in our previous paper [13]. Here, we simply emphasize some key points about the experimental setup rather than describe the details. First, a tunable laser in the 1550-nm band was over-coupled into the DPPLN microdisk with a tapered fiber. The wavelength, power and polarization of the pump could be controlled before it was launched into the tapered fiber. Second, the same tapered fiber was used to couple the nonlinear optical signals out of the microdisk cavity. The power and wavelength of the nonlinear signals were measured by a monochromator with a 0.2-nm wavelength resolution and less than 1-pW sensitivity. Third, the polarization of the scattered light of the disk cavity was confirmed by detecting the light passing through a polarization prism. The confirmation of the mode polarization helps us to assign the quantum number of the whispering gallery mode and the exact nonlinear coefficient of LN associated with nonlinear optical processes. In experiments, we first scan the wavelength of the pump laser from 1520 nm to 1570 nm with a 3-nm/s speed to roughly find the pump wavelengths with relatively large nonlinear signals. Then, we tuned the polarization of the pump light while slowly scanning the laser wavelength in a very narrow band near 1543.6 nm, where a significant nonlinear signal was initially observed, to optimize the nonlinear optical signal. It is worth mentioning that the coupling between the DPPLN disk and the tapered fiber that influences both the pump load and signal output also needs to be carefully manipulated during signal optimization.

Figure 3(a) shows the SHG observed under the pump of a 1543.6-nm wavelength and 0.02-mW power. The dependence of SHG efficiency on the pump power is shown in Fig. 3(b), which indicates a normalized conversion efficiency of  $51\% \text{ W}^{-1}$ . When the pump power was increased to 6.65 mW, we observed THG and FHG, which are shown in Figs. 3(c) and 3(d), respectively. To stabilize the nonlinear signals, the



**Table 1. Reciprocal Vectors Involved in Nonlinear Optical Processes in DPPLN Microdisks**

| Nonlinear Optical Processes (nm)        | $\Delta k$ ( $\mu\text{m}^{-1}$ ) | $G_{m,n}$ ( $\mu\text{m}^{-1}$ ) |
|-----------------------------------------|-----------------------------------|----------------------------------|
| SHG 1542.5 + 1542.5 $\rightarrow$ 771.2 | 2.46                              | 2.46                             |
| THG 1542.5 + 771.2 $\rightarrow$ 514.1  | 1.82                              | 1.88                             |
| FHG 771.2 + 771.2 $\rightarrow$ 385.6   | 6.55                              | 6.60                             |
| FHG 1542.5 + 514.1 $\rightarrow$ 385.6  | 7.19                              | 7.18                             |

coupling was kept unchanged by attaching the tapered fiber to the DPPLN disk [29], and the resonance oscillation was intentionally avoided [30]. The appearance of the THG and FHG indicates that the quasi-phase matchings for cascaded second-order effects were simultaneously fulfilled due to the abundant reciprocal wave vectors of DPPLN disks. In Figs. 3(c) and 3(d), we can read the power of THG and FHG and then derive their conversion efficiencies of  $0.1\% \text{ W}^{-2}$  and  $103\% \text{ W}^{-3}$ , respectively. The pump and SHG were confirmed to be both transverse magnetic (TM) modes with an electric field parallel to the LN optical axis by identifying the polarization of light scattered from the cavity mode being excited by lasers. By comparing the measured resonance wavelengths of the cold cavity with the theoretical ones calculated with AFM measured disk geometric parameters [13], the modes associated with the pump, SHG, THG, and FHG are approximately assigned to  $\text{TM}_{1,145}$ ,  $\text{TM}_{1,341}$ ,  $\text{TM}_{3,523}$ , and  $\text{TE}_{1,818}$ , respectively, where the first (second) number indicates the radial (azimuthal) quantum number. This means that  $d_{33}$  was utilized for SHG and THG. Accordingly, the wave-vector mismatches  $\Delta k$  for SHG, THG, and FHG are  $2.46 \mu\text{m}^{-1}$ ,  $1.82 \mu\text{m}^{-1}$ , and  $6.55 \mu\text{m}^{-1}$  ( $7.19 \mu\text{m}^{-1}$ ), respectively, as shown in Table 1. These wave-vector mismatches can be compensated for by the reciprocal vectors  $G_{15,-3} = 2.46 \mu\text{m}^{-1}$ ,  $G_{-33,9} = 1.88 \mu\text{m}^{-1}$ , and  $G_{-19,7} = 6.60 \mu\text{m}^{-1}$  ( $G_{29,-5} = 7.18 \mu\text{m}^{-1}$ ) in the DPPLN disk, which are listed in Table 1 and marked by red, blue, and purple arrows in Fig. 1(b), respectively. The normalized amplitudes of these reciprocal vectors are calculated to be 0.90%, 0.14%, 0.30% (0.27%), respectively, which are part of the reason leading to low nonlinear conversion efficiencies. Note that the resonance wavelengths of the pump and nonlinear signals in Fig. 3 are slightly different from those listed in Table 1. The reason for this is that the resonance wavelengths shown in Table 1 were measured at low pump power, while the nonlinear signals were collected at relatively high pump power leading to red shift of cavity resonance wavelengths. The sources of the side peaks of THG and FHG are unknown and will be systematically investigated in the future.

To conclude, series of coupled nonlinear optical signals were simultaneously observed in an on-chip LN microdisk cavities with a dual-period domain structure. The concurrent appearance of the second-order nonlinear optical signals is a result of the introduction of the DPPLN structure with abundant reciprocal vectors, which provide more opportunity to realize quasi-phase matching for multiple nonlinear processes. The DPPLN microdisk cavities have unit domain cells of 90-nm width, which are almost impossible to achieve in bulk LN. This work has taken an important stride towards efficient cascaded nonlinear optical effects in on-chip LN devices with a weak pump.

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## REFERENCES

1. K. J. Vahala, *Nature* **424**, 839 (2003).
2. V. S. Ilchenko, A. A. Savchenkov, A. B. Matsko, and L. Maleki, *Phys. Rev. Lett.* **92**, 043903 (2004).
3. J. Moore, M. Tomes, T. Carmon, and M. Jarrahi, *Opt. Express* **19**, 24139 (2011).
4. J. Lin, Y. Xu, Z. Fang, M. Wang, N. Wang, L. Qiao, W. Fang, and Y. Cheng, *Sci. Chin. Phys. Mech. Astron.* **58**, 114209 (2015).
5. J. Lin, Y. Xu, J. Ni, M. Wang, Z. Fang, L. Qiao, W. Fang, and Y. Cheng, *Phys. Rev. Appl.* **6**, 014002 (2016).
6. J. Moore, J. K. Douglas, I. W. Frank, T. A. Friedmann, R. M. Camacho, and M. Eichenfield, *Conference on Lasers Electro-Optics* (2016), paper STh3P.1.
7. L. Wang, C. Wang, J. Wang, F. Bo, M. Zhang, Q. Gong, M. Lončar, and Y. Xiao, *Opt. Lett.* **43**, 2917 (2018).
8. Z. Hao, L. Zhang, A. Gao, W. Mao, X. Lyu, X. Gao, F. Bo, F. Gao, G. Zhang, and J. Xu, *Sci. Chin. Phys. Mech. Astron.* **61**, 114211 (2018).
9. R. Wolf, Y. Jia, S. Bonaus, C. S. Werner, S. J. Herr, I. Breunig, K. Buse, and H. Zappe, *Optica* **5**, 872 (2018).
10. J. Lin, N. Yao, Z. Hao, J. Zhang, W. Mao, M. Wang, W. Chu, R. Wu, Z. Fang, L. Qiao, W. Fang, F. Bo, and Y. Cheng, *Phys. Rev. Lett.* **122**, 173903 (2019).
11. J. Chen, Z. Ma, Y. M. Sua, Z. Li, C. Tang, and Y. Huang, *Optica* **6**, 1244 (2019).
12. J. Lu, J. B. Surya, X. Liu, A. W. Bruch, Z. Gong, Y. Xu, and H. X. Tang, *Optica* **6**, 1455 (2019).
13. Z. Hao, L. Zhang, W. Mao, A. Gao, X. Gao, F. Gao, F. Bo, G. Zhang, and J. Xu, *Photon. Res.* **8**, 311 (2020).
14. Z. Hao, J. Wang, S. Ma, W. Mao, F. Bo, F. Gao, G. Zhang, and J. Xu, *Photon. Res.* **5**, 623 (2017).
15. R. Luo, H. Jiang, S. Rogers, H. Liang, Y. He, and Q. Lin, *Opt. Express* **25**, 24531 (2017).
16. M. Wang, N. Yao, R. Wu, Z. Fang, S. Lv, J. Zhang, L. Qiao, J. Lin, W. Fang, and Y. Cheng, "Simultaneous generation of four-wave mixing and stimulated Raman scattering in coupled lithium niobate microdisks mediated by second harmonic generation," arXiv preprint arXiv:1803.11341 (2018).
17. K. Sasagawa and M. Tsuchiya, *Appl. Phys. Express* **2**, 122401 (2009).
18. D. V. Strekalov, A. S. Kowligy, Y. Huang, and P. Kumar, *New J. Phys.* **16**, 053025 (2014).
19. X. Ye, S. Liu, Y. Chen, Y. Zheng, and X. Chen, *Opt. Lett.* **45**, 523 (2020).
20. A. Savchenkov, A. Matsko, M. Mohageg, D. Strekalov, and L. Maleki, *Opt. Lett.* **32**, 157 (2007).
21. J. Furst, D. Strekalov, D. Elser, A. Aiello, U. L. Andersen, C. Marquardt, and G. Leuchs, *Phys. Rev. Lett.* **105**, 263904 (2010).
22. T. Beckmann, H. Linnenbank, H. Steigerwald, B. Sturman, D. Haertle, K. Buse, and I. Breunig, *Phys. Rev. Lett.* **106**, 143903 (2011).
23. J. Moore, M. Tomes, T. Carmon, and M. Jarrahi, *Appl. Phys. Lett.* **99**, 221111 (2011).
24. R. Wu, J. Zhang, N. Yao, W. Fang, L. Qiao, Z. Chai, J. Lin, and Y. Cheng, *Opt. Lett.* **43**, 4116 (2018).
25. Z. Liu, S. Zhu, Y. Zhu, Y. Qin, J. He, C. Zhang, H. Wang, N. Ming, X. Liang, and Z. Xu, *Jpn. J. Appl. Phys.* **40**, 6841 (2001).
26. S. Zhu, Y. Zhu, and N. Ming, *Science* **278**, 843 (1997).
27. R. Riklund, M. Severin, and Y. Liu, *Int. J. Mod. Phys. B* **1**, 121 (1987).
28. R. Gainutdinov, T. Volk, and H. Zhang, *Appl. Phys. Lett.* **107**, 162903 (2015).
29. F. Bo, S. H. Huang, S. K. Özdemir, G. Zhang, J. Xu, and L. Yang, *Opt. Lett.* **39**, 1841 (2014).
30. J. Wang, B. Zhu, Z. Hao, F. Bo, X. Wang, F. Gao, Y. Li, G. Zhang, and J. Xu, *Opt. Express* **24**, 21869 (2016).